

Notes: Graham, Harvey, and Puri (JFE, 2013)

Managerial Attitudes and Corporate Actions

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1 Big picture

- **Question.** Do the attitudes and psychological traits of top executives help explain corporate decisions? The paper asks whether CEO optimism, risk-aversion, time preference, sure-loss aversion, and career characteristics are related to capital structure, acquisitions, growth, and compensation.
- **Setting.** The authors use survey responses from senior executives, with the core analysis focusing mainly on US CEOs. The survey also includes US CFOs and non-US executives, which allows the paper to compare CEOs with CFOs and US executives with non-US executives.
- **Core measurement idea.** Instead of inferring executive traits from observed firm actions, the paper directly measures attitudes using survey questions and established psychometric tools from economics and psychology.
- **Main comparison result.** CEOs are much less risk-averse and much more optimistic than the lay population. CEOs also differ from CFOs: CEOs are more optimistic, older, have longer tenure, and hold much larger ownership stakes.
- **International comparison.** US CEOs and CFOs are more risk-tolerant, more optimistic, and more patient than their non-US counterparts. This provides a potential channel for cross-country differences in corporate policies.
- **Capital structure result.** Firms led by very optimistic CEOs are more likely to use debt, especially short-term debt. This is consistent with theories in which optimistic managers expect favorable future outcomes and therefore are more comfortable with short-maturity obligations.
- **M&A result.** More risk-tolerant CEOs are more likely to make acquisitions. In the logit specification, a highly risk-averse CEO is associated with a substantially lower probability of acquisition activity.
- **Growth and matching result.** Risk-tolerant, younger, taller, and non-finance-background CEOs are more likely to be found in high-growth firms. The paper interprets this as consistent with either CEO traits affecting growth or firms and managers matching on traits.
- **Compensation result.** Risk-averse CEOs receive less performance-sensitive compensation and more salary. Impatient CEOs also receive less stock/options/bonus compensation. This provides direct evidence consistent with agency theory: incentive pay is costlier when the manager is more risk-averse or discounts the future heavily.
- **Economic takeaway.** Managerial attitudes are not just background personality details. Directly measured CEO traits are systematically related to corporate policies and to the design of executive compensation contracts.

2 Incremental contribution of the paper

- **From indirect proxies to direct measurement.** Earlier behavioral corporate finance papers often infer overconfidence or optimism from actions such as option exercise or acquisition decisions. This paper directly asks executives questions designed to measure attitudes.
- **First direct psychometric evidence on sitting executives.** The paper administers psychometric-style questions to senior executives, including CEOs and CFOs. This is unusual because executive traits are normally unobservable in standard archival data.

- **CEO-versus-CFO comparison.** The paper shows that CEOs and CFOs differ not only in job function and career path but also in measured attitudes. This helps clarify what may be special about people who become chief executives.
- **US-versus-non-US comparison.** The paper documents that US executives differ from non-US executives in risk tolerance, optimism, patience, and sure-loss aversion. This adds a behavioral channel to discussions of why firms behave differently across countries.
- **Link from traits to corporate policy.** The paper connects measured executive traits to standard corporate finance decisions: leverage, debt maturity, and acquisitions.
- **Link from traits to firm matching.** The paper asks whether executives with particular traits are found in particular types of firms, especially high-growth firms.
- **Compensation-contract contribution.** The paper provides rare direct evidence that risk-aversion and time preference are related to the mix of salary and performance-sensitive pay. This is important because standard agency theory predicts such a relation, but direct empirical evidence had been limited.
- **Private-firm contribution.** The sample includes many private firms, which are usually difficult to study in corporate finance. This broadens the evidence beyond the public-firm samples commonly available in Compustat.
- **Governance implication.** Boards, investors, and compensation committees should recognize that managerial attitudes can be measured and that those attitudes are related to financing, acquisition, and pay-design outcomes.

3 Data and measurement

3.1 Survey sample and delivery

- The survey was sent to several groups of CEOs, including subscribers to *Chief Executive* magazine, CEO readers of *CFO* magazine, and executives associated with the World Economic Forum in Davos.
- After invalid addresses were removed, roughly 10,700 CEOs were surveyed. The authors received 1,180 CEO responses, for a response rate around 11%.
- The main CEO analysis focuses primarily on 1,017 CEOs whose firms are headquartered in the United States.
- The authors also surveyed US CFOs, European CFOs, and Asian CFOs. These groups provide comparison samples for executive characteristics and attitudes.
- Because the survey is anonymous, the authors cannot match most responses to external archival firm identifiers, but anonymity likely helps elicit more candid responses.

Interpretation: The survey approach gives the paper access to traits and beliefs that are normally unobservable. The cost is that the data are self-reported, cross-sectional, and mostly anonymous.

3.2 Measuring attitudes

- **Risk-aversion.** The authors adapt the Barsky et al. lifetime-income gamble questions. Executives choose between a safe job paying current income for life and risky jobs with equal chances

of higher and lower lifetime income. Respondents who reject even the safer risky gamble are classified as highly risk-averse.

- **Optimism.** The authors use the Life Orientation Test-Revised, or LOT-R, a standard psychology instrument. Six scored questions are coded so that higher values indicate greater dispositional optimism. Executives with an average score of 3 or higher on a 0–4 scale are classified as very optimistic.
- **Time preference.** Executives choose between receiving \$10,000 now and \$13,000 one year later. Those preferring the immediate payment are classified as impatient, implying a discount rate above 30%.
- **Aversion to sure losses.** Executives decide how much they would invest in a previously failed project with only a 20% chance of generating a future payoff. A willingness to invest \$2 million or more is classified as aversion to sure losses.

Interpretation: These measures are designed to capture stable behavioral tendencies rather than to infer attitudes from the corporate policies that the paper later studies.

3.3 Executive demographics and career variables

- The survey records whether the executive has an MBA, gender, age, height, tenure, university SAT score, and stock ownership.
- The authors also record whether the executive’s prior experience was focused in finance or accounting.
- Height is used partly because prior literature links height to confidence, status, and potentially cognitive or social advantages.
- Gender is included because prior work suggests that men and women may differ in overconfidence and risk-related behavior.
- MBA status, elite education, age, and career path proxy for training, selection, experience, and managerial background.

Interpretation: These variables help separate psychological traits from more standard resume characteristics. They also allow the authors to ask whether firm policies are related to attitudes after controlling for observable background.

3.4 Firm characteristics and corporate policies

- The survey records whether the firm is public, firm size, the number of operating segments, historical sales growth, and expected future sales growth.
- Capital structure variables include the debt-to-assets ratio and the short-term debt-to-total-debt ratio.
- Acquisition activity is measured by the number of acquisitions over the previous two years.
- Compensation is measured as target percentages of total pay coming from salary, bonus, stock and options, and other forms of compensation.
- The main compensation variables are salary share and performance-sensitive compensation share, where performance-sensitive compensation combines stock, options, and bonuses.

Interpretation: The survey collects corporate-policy variables directly because anonymity prevents systematic matching to Compustat. The authors later benchmark the public-firm subset against Compustat to assess representativeness.

4 Models and empirical specifications

4.1 Group comparisons

The paper first compares means across executive groups:

- US CEOs versus US CFOs;
- US CEOs versus non-US CEOs;
- US CFOs versus non-US CFOs;
- CEOs versus external population benchmarks for risk-aversion and optimism.

The comparisons are reported both unconditionally and after matching by public/private status or firm-size category.

Interpretation: These comparisons establish that executives, especially CEOs, differ strongly from other groups. They do not by themselves identify causality, but they motivate the firm-policy regressions.

4.2 Capital-structure specifications

For leverage and debt maturity, the paper estimates models of the form

$$Policy_i = \alpha + \beta_1 RiskAverse_i + \beta_2 Optimistic_i + \beta_3 SureLoss_i + \beta_4 Impatient_i + \Gamma' X_i + \delta_{industry} + \varepsilon_i,$$

where $Policy_i$ is either an indicator for a high debt ratio or an indicator for a high short-term-debt ratio.

- The four key attitude variables are highly risk-averse, very optimistic, averse to sure losses, and impatient.
- Controls include gender, height, MBA, finance/accounting background, age, public status, prestigious college, large firm, and past sales growth.
- Industry dummy variables are included.
- The regressions are interpreted as linear probability associations for high-debt and high-short-term-debt outcomes.

Interpretation: The capital-structure regressions ask whether measured traits are related to leverage and debt maturity after controlling for basic executive and firm characteristics.

4.3 Acquisition specifications

For acquisition activity, the paper estimates logit, ordered logit, and OLS specifications:

$$Acquisition_i = \alpha + \beta_1 RiskAverse_i + \beta_2 Optimistic_i + \beta_3 SureLoss_i + \beta_4 Impatient_i + \Gamma' X_i + \delta_{industry} + \varepsilon_i.$$

- One dependent variable is an indicator for whether the firm made any acquisitions.
- Another specification distinguishes no acquisition, one acquisition, and more than one acquisition.
- A third specification uses $\ln(Acq + 1)$ as a continuous acquisition measure.
- The acquisition regressions focus on CEOs with at least two years of tenure because the acquisition window covers the previous two years.

Interpretation: The key hypothesis is that risk-tolerant CEOs are more willing to undertake acquisitions, consistent with acquisition activity involving managerial risk-taking and possible hubris.

4.4 Growth and compensation specifications

For firm growth and compensation, the paper estimates logit models such as

$$Outcome_i = \alpha + \beta_1 RiskAverse_i + \beta_2 Optimistic_i + \beta_3 SureLoss_i + \beta_4 Impatient_i + \Gamma' X_i + \delta_{industry} + \varepsilon_i.$$

- Growth outcomes include high expected sales growth and high historical sales growth.
- Compensation outcomes include high stock/options/bonus compensation and high salary compensation.
- The compensation regressions include executive stock ownership as an additional control.
- Marginal effects are reported to translate logit coefficients into changes in outcome probabilities.

Interpretation: The compensation analysis connects directly to agency theory. If incentive pay imposes risk on the CEO, more risk-averse executives should require more fixed compensation and less performance-sensitive compensation.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies robust correlations between directly measured executive traits and firm policies.
- The design is strongest for showing that traits are economically meaningful and are related to real corporate decisions.
- It is not a randomized design, so the paper does not prove that traits causally determine policies.
- The results can be interpreted as evidence of either managerial influence, sorting of managers into firms, or both.

Interpretation: The safest reading is predictive and diagnostic: executive attitudes are systematically related to corporate actions. The evidence is not a clean causal estimate of changing a CEO's optimism or risk tolerance.

5.2 Selection, matching, and reverse causality

- Firms may select CEOs with certain traits because those traits fit the firm’s existing policy or strategy.
- CEOs may self-select into firms that match their own preferences, such as risk-tolerant CEOs joining high-growth companies.
- Firm policies may shape attitudes over time, although the authors cite evidence that optimism tends to be persistent.
- Because the survey is cross-sectional, the paper generally cannot separate CEO fixed effects from firm fixed effects.

Interpretation: The matching interpretation is not a weakness for every question. For compensation and growth, matching between CEO type and firm type is itself economically meaningful. For causal statements about policy choices, however, it remains a limitation.

5.3 Measurement advantages and limitations

- The main measurement advantage is direct observation of attitudes that are usually latent.
- The survey uses established measurement approaches for risk-aversion and optimism.
- The survey includes many private firms, broadening the evidence beyond public companies.
- The main measurement limitations are survey nonresponse, self-reported firm variables, possible misunderstanding of questions, and the inability to match most firms to external data.
- The authors benchmark the public-firm subset against Compustat and find that self-reported leverage is reasonably consistent with Compustat for identifiable public firms.

Interpretation: The paper’s credibility comes from combining unusual direct attitude data with standard corporate-finance outcomes. Its main empirical caution is that all relations are observational and rely heavily on survey measurement.

6 Tables and figures

6.1 Table 1: Firm characteristics and Table 2: CEO personal characteristics

Read:

- Table 1 shows that the sample is dominated by private firms: only 11.5% of sample firms are public.
- The mean sales revenue is \$551 million, while the median is only \$62 million, indicating a right-skewed firm-size distribution.
- The median debt-to-assets ratio is 20%, and the median short-term-debt-to-total-debt ratio is 11%.
- Public firms are much larger, have more performance-sensitive compensation, and have lower CEO stock ownership than the full sample.
- Table 2 shows that only 9.8% of US CEOs are highly risk-averse, while 80.2% are very optimistic.

- CEOs are predominantly male, about one-third have MBA degrees, and roughly one-third are classified as impatient.

Economic message: Table 1 defines the firm-policy environment, and Table 2 defines the key executive traits. The striking starting point is that CEOs look very different from the general population: they are unusually risk-tolerant and optimistic.

Table 1

Firm characteristics summary statistics.

This table reports the summary statistics of specific firm characteristics used in the analysis. All variables reflect the survey responses of participating US CEOs. Firm size (size) is determined based on survey question categorizing firm sales into seven size groups. Category midpoints are used to create the firm-size variable reported here. Category 1 includes firms with sales less than \$25 M, category 2 includes firms with sales between \$25 M and \$99 M, category 3 includes firms with sales between \$100 M and \$499 M, category 4 includes firms with sales between \$500 M and \$999 M, category 5 includes firms with sales between \$1B and \$4.9B, category 6 includes firms with sales between \$5B and \$9.9B, and category 7 includes firms with \$10B or above in sales. Historical and expected growth reflect growth in sales. Number of operating segments refers to the number of segments in which the reporting executive has experience in prior to current position. Number of acquisitions reflects the number of acquisitions the firm has made over the past two years. Executive compensation variables (stocks, options, bonuses, and salary) are percentages of total compensation.

	Observations	Mean	Std. dev.	Minimum	Median	Maximum
<i>Panel A: All firms</i>						
Public (%)	785	11.5	N/A	N/A	N/A	N/A
Sales (size) revenue (\$M)	961	551.0	1,819.3	12.5	62	12,500
# Operating segments	1,001	2.2	1.4	1.0	2	12
Historical growth (%)	948	22.2	45.4	-30.0	12	500
Expected growth (%)	952	18.6	19.2	-5.0	12	99
Debt-to-assets ratio (%)	811	26.6	25.2	0.0	20	100
Short-term Debt to Total Debt Ratio (%)	805	25.0	29.6	0.0	11	100
Number of acquisitions	872	0.7	1.5	0.0	0	9
Stocks, options, and bonuses (%)	793	36.7	26.2	0.0	35.0	100
Salary (%)	795	57.1	26.4	0.0	60.0	100
Stock ownership (%)	788	33.9	36.5	0.0	17.5	100
<i>Panel B: Public firms only</i>						
Public (%)	90	100.0	N/A	N/A	N/A	N/A
Sales (size) revenue (\$M)	90	2,378.6	3,810.4	12.5	299.5	12,500
# Operating segments	88	2.5	1.3	1.0	3	6
Historical growth (%)	90	15.2	18.9	-10.0	8.5	125
Expected growth (%)	90	13.5	12.5	1.0	10	75
Debt-to-assets ratio (%)	79	28.3	26.1	0.0	25.0	100
Short-term debt to total debt ratio (%)	79	20.5	25.5	0.0	10.0	100
Number of acquisitions	88	1.2	1.6	0.0	1.0	9
Stocks, options, and bonuses (%)	87	52.4	23.3	0.0	55.0	98
Salary (%)	87	44.6	22.9	0.0	40.0	100
Stock ownership (%)	88	5.4	14.80	0.0	1.0	100

(a) Table 1: Firm characteristics summary statistics

Table 2

Personal characteristics of corporate executives.

This table reports the summary statistics of personal characteristics of US CEOs. Highly risk-averse is a dummy variable for executives that prefer a job that pays with certainty their current salary against jobs that pay twice their current income with 50% chance and 80% of their current income with 50% chance. MBA degree is a dummy variable for executives with MBA degrees. Very optimistic is a dummy variable for an executive that has reported an average of 4 or higher in the LOT-R Test. Averse to sure losses is a dummy for executives who avoids sure losses now by hedging. Focused in fin. & acc. is a dummy for executives who have experience focused in financial and/or accounting fields. Tenure is the number of years the executive has been in current position. Impatient is a dummy variable for executives who prefer money now over money a year later at an interest rate of 30%. Stock ownership measures the fraction of a firm's stock that the executive will own if all stock options are exercised.

Summary statistics						
	Observations	Mean	Std. dev.	Minimum	Median	Maximum
Highly risk-averse (%)	1,008	9.8	29.8	0	0	100
MBA Degree (%)	916	34.9	47.7	0	0	100
Male (%)	1,009	92.3	26.7	0	100	100
Male Height (inches)	881	71.2	2.8	52	71	82
Female Height (inches)	78	65.7	2.4	58	66	71
Very Optimistic (%)	992	80.2	39.8	0	100	100
Averse to sure losses (%)	861	8.5	27.9	0	0	100
Impatient (%)	986	32.9	47.0	0	0	100
Focused in Fin. & Acc. (%)	886	16.0	36.7	0	0	100
Age	992	54.1	9.4	25	54	89
University SAT Score	842	1,159.8	156.6	750	1,140	1,485
Tenure	1,011	10.4	8.4	0.5	8	96
Stock Ownership (%)	788	33.9	36.5	0	17.5	100

Benchmarking this against the norms in the psychology (4.33). Of our CEOs, 80% are very optimistic (mean score of

(b) Table 2: Personal characteristics of corporate executives

6.2 Table 3: Comparison of executive characteristics across samples

Read:

- Panel A compares US CEOs with US CFOs. CEOs are much more likely to be very optimistic, less likely to have MBAs, older, more likely to have longer tenure, and much more likely to own firm stock.
- The difference in career path is large: only 15.6% of US CEOs are finance/accounting focused, compared with 86.7% of US CFOs.
- Panel B compares US CEOs with non-US CEOs. US CEOs are less risk-averse, much more optimistic, less averse to sure losses, and less impatient.
- Panel C shows similar international patterns among CFOs: US CFOs are less risk-averse, more optimistic, and more patient than non-US CFOs.
- The paper repeats comparisons after public/private matching and size matching. The main attitude differences remain visible.

Economic message: Table 3 establishes that top executives are not behaviorally homogeneous. CEO status and country are associated with large differences in measured attitudes, which provides a plausible behavioral channel for differences in corporate behavior.

Table 3

Comparison of executive characteristics across samples.

This table reports the summary statistics of specific personal characteristics of participating executives used in the analysis. Panels A-C compare the characteristics of US CEOs against those of US CFOs, US CEOs with Non-US CEOs and US CFOs with Non-US CFOs, respectively, based on three sampling methods. The first, unconditional, uses the entire sample. The second sampling method matches firms one-on-one based on public/private status, randomly sampling when the number of observations are uneven. The third sample method matches firms one-on-one based on size categories. Panel A compares US CEOs to US CFOs using the three sampling methods described above, while Panel B compares the characteristics of US CEOs against those of non-US CEOs in the same way. Panel C compares the characteristics of US CFOs against those of non-US CFOs for the three sampling methods. Variables are defined as in Table 2. Summary statistics reflect means where applicable. Significance at the 10% level is denoted by *, 5% by **, and 1% by ***. The number of observations is included in the panel header.

Panel A: Comparisons between US CEOs and US CFOs based on three sampling methods

	(1) Unconditional~1011 US CEOs vs US CFOs		(2) Public/private matching~383 US CEOs vs US CFOs		(3) Size matching~417 US CEOs vs US CFOs	
Highly risk-averse (%)	9.8	8.4	10.8	7.0*	10.6	8.7
MBA degree (%)	34.9	51.8***	34.1	49.3***	32.8	49.5***
Male (%)	92.3	90.4	93.7	93.0	93.7	90.1***
Male height (inches)	71.2	71.1	70.9	71.2	70.9	
Female height (inches)	65.7	65.1	66.2	64.5**	66.4	64.7**
Very optimistic (%)	80.2	65.3***	83.5	65.7***	79.7	66.3***
Averse to sure losses (%)	8.5	13.2***	8.8	11.7	7.8	11.6*
Impatient (%)	32.8	26.4***	30.4	24.9*	28.0	26.8
Focused in fin. & acc. (%)	15.6	46.7***	15.9	85.2***	16.8	87.7***
Age	54.1	48.6***	54.0	49.0***	55.0	48.7***
University SAT score	1159.8	1113.5***	1165.8	1109.0***	1145.2	1106.3***
Tenure (years)	10.4	6.8***	9.9	7.1***	10.3	6.9***
Stock ownership (%)	33.9	9.0***	33.6	4.0***	31.7	6.4***

(a) Table 3, Panel A: US CEOs versus US CFOs

Tenure (years)	10.4	6.8***	9.9	7.1***	10.3	6.9***
Stock ownership (%)	33.9	9.0***	28.7	4.5***	20.0	3.8***

Panel B: Comparisons between US CEOs and non-US CEOs based on three sampling methods

	(1) Unconditional~1011 US CEOs vs Non-US CEOs		(2) Public/private matching~120 US CEOs vs Non-US CEOs		(3) Size matching~153 US CEOs vs Non-US CEOs	
Highly risk-averse (%)	9.8	16.7***	5.8	17.6***	9.2	17.1**
MBA degree (%)	34.9	48.2***	33.9	51.9***	39.9	48.9
Male (%)	92.3	94.4	92.5	95.0	94.7	94.7
Male height (inches)	71.2	69.3***	70.9	69.7***	71.1	69.2***
Female height (inches)	65.7	56.9***	67.1	53.1	66.8	55.7
Very optimistic (%)	80.2	54.1***	80.7	57.6***	78.7	53.1***
Averse to sure losses (%)	8.5	19.2***	4.3	17.9***	8.6	19.3***
Impatient (%)	32.8	41.4**	28.3	38.7*	36.2	41.1
Focused in fin. & acc. (%)	15.6	25.3***	18.3	26.9	19.0	25.0
Age	54.1	50.0***	54.9	49.7***	52.9	49.7***
University SAT score	1159.8	1208.7	1154.2	1189.6	1157.5	1201.5
Tenure (years)	10.4	9.0*	10.0	8.0*	10.3	8.5*
Stock ownership (%)	33.9	33.6	31.7	33.4	40.2	33.6

(b) Table 3, Panel B: US CEOs versus non-US CEOs

University SAT score	1159.8	1208.7	1154.2	1189.6	1157.5	1201.5
Tenure (years)	10.4	9.0*	10.0	8.0*	10.3	8.5*
Stock ownership (%)	33.9	33.6	31.7	33.4	40.2	33.6

Panel C: Comparisons between US CFOs and non-US CFOs based on three sampling methods

	(1) Unconditional~534 US/707 Non-US CFOs vs Non-US CFOs		(2) Public/private matching~431 US CFOs vs Non-US CFOs		(3) Size matching~430 US CFOs vs Non-US CFOs	
Highly risk-averse (%)	8.4	13.9***	7.1	13.9***	7.7	14.9***
MBA degree (%)	51.8	33.6***	52.0	31.7***	52.3	34.3***
Male (%)	90.4	87.6	92.6	90.0	90.2	89.3
Male height (inches)	71.0	69.1***	70.9	68.7***	71.0	69.2***
Female height (inches)	65.1	63.4	64.9	64.4	65.1	63.9**
Very optimistic (%)	65.3	47.9***	65.1	47.7***	64.9	51.6***
Averse to sure losses (%)	13.2	13.9	13.3	14.1	13.6	12.8
Impatient (%)	26.4	37.9***	24.5	38.8***	26.8	35.3***
Focused in fin. & acc. (%)	86.7	85.3	85.5	85.2	86.4	87.0
Age	48.6	43.3***	48.8	43.5***	48.8	43.5***
University SAT score	1113.5	1118.8	1117.6	1100.0	1115.1	1093.0
Tenure (years)	6.8	5.9***	6.9	5.8***	6.7	5.5***
Stock ownership (%)	3.8	6.1**	4.0	6.4*	3.9	5.3

Table 3, Panel C: US CFOs versus non-US CFOs.

6.3 Table 4: Correlation table

Read:

- Older CEOs are significantly more likely to be highly risk-averse.
- MBA status is positively correlated with a finance/accounting background.
- Public status is strongly correlated with being a large firm.
- CEO stock ownership is negatively correlated with public-firm status and large-firm status.
- Historical growth is negatively correlated with CEO age and positively related to some characteristics associated with confidence or growth-firm matching.

Economic message: Table 4 shows that traits, demographics, and firm characteristics are correlated but not redundant. This motivates multivariate regressions that include several executive and firm controls simultaneously.

Table 4

Correlation Table.
This table reports the correlations between both firm and executive characteristics for our main analysis sample of US CEOs. Old is a dummy variable for executives older than the median of the survey executives. Tenure is considered long if the executive has been in the position for longer than the median length of office in the sample of US CEOs. Prestigious college is a dummy for executives who studied at a US university with average SAT scores in the fourth quartile of the survey sample. Large firm is a dummy for firms reported to have more than \$1B in sales. Tall is a dummy variable for male executives over 5 feet 10 inches and female executives over 5 feet 4 inches. Other variables are defined as in Table 2. Significance at the 10% level is denoted by *, 5% by **, and 1% by ***.

	Highly risk-averse	Very optimistic	Averse to sure losses	Impatient	MBA	Male	Tall	Focused in fin. & acc.	Old	Public	Prestigious college	Prestigious college	Historical growth
Very optimistic	-0.013												
Averse to sure losses	0.002	0.038											
Impatient	0.038	-0.017	0.083**										
MBA	-0.041	0.010	-0.073**	-0.050									
Male	-0.004	-0.047	-0.006	-0.093***	0.048								
Tall	-0.031	0.034	0.036	0.013	0.020	0.006							
Focused in fin. & acc.	0.036	-0.036	-0.014	-0.082***	0.172***	0.064**	-0.020						
Old	0.124***	0.014	-0.088**	-0.040	-0.139***	0.061**	-0.013	-0.051					
Public	0.015	0.001	0.007	-0.041	0.005	-0.000	0.000	-0.012	0.020				
Prestigious college	-0.011	-0.014	-0.027	-0.096	0.042	-0.119***	0.003	-0.040	-0.075**	0.007			
> \$1B Firm	-0.006	0.021	0.007	-0.011	-0.024	0.037	-0.015	0.005	0.050	0.387***	-0.042		
Historical growth	-0.024***	0.005	0.035	0.058	0.065	-0.019	0.023***	-0.051**	-0.153***	-0.062***	-0.026	-0.056*	
Stock ownership	0.046	0	-0.005	0.08**	0.042	-0.076*	0.041	-0.053	0.002	-0.291***	0.107***	-0.195***	0.109***

Table 4: Correlations between firm and executive characteristics.

6.4 Table 5: CEO and firm characteristics, capital structure, and acquisitions

Read:

- Panel A reports univariate relations between executive traits and corporate policies.
- CEOs with finance/accounting backgrounds are more likely to run firms with high debt ratios.
- Very optimistic CEOs and male CEOs are more likely to run firms with high short-term-debt ratios.
- Public and large firms are much more likely to make acquisitions.
- Panel B shows the multivariate regressions. Very optimistic CEOs are associated with higher debt and higher short-term debt.
- Highly risk-averse CEOs are much less likely to make acquisitions. The marginal-effect estimate implies roughly a 24.6 percentage point lower probability of making an acquisition.
- The ordered-logit and OLS acquisition specifications tell the same story: risk-tolerant CEOs make more acquisitions.

Economic message: Table 5 is the paper's core corporate-policy result. Optimism is linked to debt maturity and leverage, while risk tolerance is linked to acquisition activity. These are exactly the kinds of policies where CEO beliefs and risk preferences could matter.

Table 5
Do CEO and/or firm characteristics affect corporate policies?

Panel A looks at how certain executive and firm characteristics affect corporate policies on capital structure and M&A in the sample of US CEO survey respondents. Debt ratio and short-term debt are defined to be high if they are above the mean of the survey sample of US CEOs. Mergers and acquisitions reflect the merger and acquisition behavior within the last two years for firms which have US CEO survey representation. Acquisitions = 1 is an indicator for firms with M&A activity. Numbers in each column reflect the percentage of US CEOs, among the ones displaying / not displaying the trait at the top, who also reflect the characteristics at the left. For example, of those with available data, 41% of males and 28% of females were associated with high debt ratios. Other variables are defined as in Table 2. Significance is between the sample that exhibits the characteristic trait and the sample that does not exhibit the trait. Significance at the 10% level is denoted by *, 5% by **, and 1% by ***.

Panel B reports the regression analysis of firm/CEO characteristics on corporate decisions with respect to M&A and capital structure for CEOs with at least 1 year of tenure. Specification 3 uses a binary dependent variable and therefore uses both a logit and a marginal effect specification to calculate the contribution of each firm/CEO characteristic to the probability of the left-hand side event. Specification 4 uses a categorical variable on acquisitions that distinguishes firms with no acquisitions (acqcat=0), one acquisition (acqcat=1), and more than one acquisition (acqcat=2), and runs an ordered logit model. Industry dummy variables are used. The ten industry classifications used in the survey are: Retail (Wholesale, Mining/Construction, Manufacturing, Transportation/Energy, Communications/Health, Tech, Banking, Service/Consulting, Healthcare, and Others). For debt regressions, debt ratio and short-term debt are defined to be high if they are above the mean of the survey sample of US CEOs. Only US CEOs are included in the regressions. All specifications include the following controls: binary indicator variables for gender, tenure, whether the CEO has an MBA, finance/accounting background, age, whether the firm is public, whether the CEO went to a prestigious college, firm size > \$1 B, and past three years revenue growth of the firm. A constant term is also included in all relevant specifications. Significance at the 10% level is denoted by *, 5% by **, and 1% by *** with z-statistics reported in parentheses.

	Highly risk-averse		Very optimistic		Averse to sure losses		Impatient		MBA		Gender		Tall		Fin / acc focused		Old		Firm type		Prestigious college		Small firm	
	High	Low	Yes	No	Yes	No	Yes	No	Yes	No	Male	Female	Yes	No	Yes	No	Yes	No	Public	Private	Yes	No	Yes	No
High debt ratio # Observations	33	41	41	35	38	40	38	41	39	39	41	39*	39	42	52	37***	39	40	47	40	39	42	39	51*
High short-term debt # Observations	26	34	35	24***	28	33	34	33	30	34	34	17***	32	36	36	32	35	30	25	34*	29	35	33	28
Acquisitions=0 # Observations	74	67*	65	66	56	67*	65	66	68	65	66	65	66	65	70	65	66	66	47	67***	71	67	68	40***
Acquisitions=1 # Observations	809	857	840	857	840	857	840	857	840	857	840	857	840	857	840	857	840	857	765	728	728	728	807	807
Acquisitions=2 # Observations	809	857	840	857	840	857	840	857	840	857	840	857	840	857	840	857	840	857	765	728	728	728	807	807

Panel B: Regression Analysis of corporate policies on executive traits

	OLS		Logit		Ordered Logit		OLS	
	1	2	3	4	5	6	7	
All CEOs for debt regressions								
High debt ratio								
High short-term debt ratio								
Made acquisitions								
No acquisitions vs. 1 acquisition vs. more than 1 acquisition								
Ln (Acq+1)								
Coefficients								
Highly risk-averse	0.004	-0.016	-1.388**	-0.246**	-1.160**	-0.257**	-0.275**	
Very optimistic	0.009*	0.000	0.000	0.000	0.000	0.000	0.000	
Averse to sure losses	0.043	0.045	0.086	0.085	0.091	0.089	0.096	
Impatient	0.029	0.016	-0.173	-0.039	-0.137	-0.040	-0.043	
	(1.10)	(1.14)	(0.76)	(0.76)	(0.63)	(0.63)	(0.63)	

(a) Table 5, Panel A: Univariate traits and corporate policies

(b) Table 5, Panel B: Regression results, main coefficients

5 (continued)

Panel B: Regression Analysis of corporate policies on executive traits

	OLS		Logit		Ordered Logit		OLS
	1	2	3		4		5
	All CEOs for debt regressions		CEO Tenure at least 2 years for acquisition regressions				
	High debt ratio	High short-term debt ratio	Made acquisitions		No acquisitions vs. 1 acquisition vs. more than 1 acquisition		Ln (Acq + 1)
	Coefficients	Coefficients	Coefficients	Marginal effects	Coefficients	Marginal effects (acqcat = 1/acqcat = 2)	Coefficient
Control variables	Yes	Yes	Yes		Yes		Yes
Industry dummies	Yes	Yes	Yes		Yes		Yes
N	507	503	504		504		504
df for Logit (R-squared for OLS)	11.94%	7.26%	39.82 (21)		50.78 (21)		13.05%
F-value	< 0.01	< 0.01	< 0.01		< 0.01		< 0.01
Pseudo R-squared	11.94%	7.26%	8.1%		7.30%		13.05%

Table 5, Panel B continued: Controls and model statistics.

6.5 Table 6: CEO traits, firm growth, and compensation

Read:

- Panel A shows univariate relations for growth and compensation outcomes.
- Risk-tolerant CEOs are more likely to work in high historical-growth and high expected-growth firms.

- Tall, younger, and non-finance-background CEOs are more common in high-growth firms.
- Risk-tolerant CEOs are more likely to receive high stock/options/bonus compensation and less likely to receive high salary compensation.
- Panel B reports multivariate regressions. Highly risk-averse CEOs are less likely to be in high expected-growth firms.
- Highly risk-averse CEOs are much less likely to receive high stock/options/bonus compensation and much more likely to receive high salary compensation.
- Impatient CEOs are also less likely to receive high stock/options/bonus compensation.

Economic message: Table 6 supports both a matching interpretation and an agency-theory interpretation. Risk-tolerant executives are found in growth firms and receive more performance-sensitive compensation; risk-averse executives receive more fixed salary.

Do CEO and/or firm characteristics affect corporate policies?

Panel A looks at how certain executive and firm characteristics affect corporate policies in terms of firm sales growth and managerial compensation in the sample of US CEO survey respondents. Historical growth is high if it is above 5% and expected growth is high if it is above 6%. Stock, options, and bonus are considered high if above the mean of all responses. Numbers in each column reflect the percentage of US CEOs, among the ones displaying not displaying the trait at the top, who also reflect the characteristic at the left (example provided in Table 5). Significance is between the sample that exhibits firm/managerial trait and the sample that does not exhibit the trait. Significance at the 10% level is denoted by a *, 5% by **, and 1% by ***.

Panel A: Performance traits and corporate policies

	Highly risk-averse		Very optimistic		Averse to sure losses		Impatient		MBA		Gender		Tall		His / acc focused		Old		Firm type		Prestigious college		Small firm	
	High	Low	Yes	No	Yes	No	Yes	No	Yes	No	Male	Female	Yes	No	Yes	No	Yes	No	Public	Private	Yes	No	Yes	No
High historical growth	64	37***	76	76	76	76	77	75	78	74	76	76	79	68***	66	78**	71	83***	66	78**	68	78***	76	72
# Observations	945	933	847	945	862	946	942	944	946	942	942	944	946	942	944	944	944	944	944	944	944	945	945	
High expected growth	62	81***	80	73*	80	78	82	77*	83	76**	79	71	88	75*	72	80*	74	85***	66	81***	78	78	80	63***
# Observations	968	937	851	948	864	950	945	931	948	945	945	945	945	945	945	945	945	945	945	945	945	945	945	
High stock, options, bonus	35	53***	51	51	63	59	42	56***	53	50	52	32***	54	40**	56	50	48	57***	81	59***	52	50	52	72***
# Observations	1088	861	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	996	
High salary	69	51***	54	48	48	53	58	50***	50	53	52	60*	52	53	44	54**	56	47**	30	56***	44	50***	55	35***
# Observations	793	781	784	792	723	794	791	779	792	749	600	790												

	1 High expected growth		2 High stock, options, and bonus		3 High salary	
	Coefficients	Marginal effects	Coefficients	Marginal effects	Coefficients	Marginal Effects
Highly risk-averse	-0.804*	-0.080	-0.900**	-0.223**	1.152***	0.287***
Very optimistic	0.421	0.040	0.027	0.007	0.126	0.031
Averse to sure losses	0.207	0.002	0.527	0.131	-0.336	-0.079
Impatient	0.282	0.003	-0.432**	-0.112**	0.186	0.041
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	513.0	506.0	513.0	506.0	513.0	506.0
Adjusted R-squared	0.148	0.148	0.148	0.148	0.148	0.148
p-value	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Pseudo R-squared (2)	38.5	38.2	38.2	38.2	38.5	38.5

(a) Table 6, Panel A: Growth and compensation, univariate patterns

(b) Table 6, Panel B: Growth and compensation regressions

6.6 Appendix Tables A1 and A2: Benchmarking to Compustat

Read:

- Table A1 compares the full US-CEO survey sample with Compustat breakpoints using survey-variable definitions.
- The full sample is smaller than the typical Compustat firm, which is expected because the survey contains many private firms.
- Table A2 repeats the comparison for the public-firm subset.
- The public-firm subset is more similar to Compustat, especially for debt/assets, debt maturity, and historical growth.

Economic message: The appendix benchmarking is important because survey samples are often criticized for representativeness. The paper shows that although the full sample is smaller than Compustat, the public-firm subset is reasonably comparable on key financial variables.

Table A1

Comparison of the survey sample of the firms with US CEOs to the Compustat sample.

The survey data are matched against the 2005 annual Compustat data for consistency as our survey is conducted in early 2006, which should reflect 2005 financials. Sales is surveyed by seven size categories. Category 1 includes firms with sales less than \$25 M, category 2 includes firms with sales between \$25 M and \$99 M, category 3 includes firms with sales between \$100 M and \$499 M, category 4 includes firms with sales between \$500 M and \$999 M, category 5 includes firms with sales between \$1B and \$4.9B, category 6 includes firms with sales between \$5B and \$9.9B, and category 7 includes firms with \$10B or above in sales. Debt-to-assets ratio, short-term debt ratio, dividend yield, and historical growth are reported by US CEO survey respondent.

Credit rating is the Standard and Poors (S&P) long-term domestic issuer credit rating grouped into ten categories in order of increasing credit ratings. Category 1 includes firms with credit ratings of D or suspended, category 2 includes firms with ratings of C, or CI, category 3 includes firms with ratings of CC, category 4 includes firms with ratings of CCC-, CCC, or CCC+, category 5 includes firms with ratings of B-, B, or B+, category 6 includes firms with ratings of BB-, BB, or BB+, category 7 includes firms with ratings of BBB-, BBB, or BBB+, category 8 includes firms with ratings of A-, A, or A+, category 9 includes firms with ratings of AA-, AA, or AA+, and category 10 includes firms with ratings of AAA- or AAA.

Variable	Sample average	Sample median	Compustat breakpoint quintiles				
			1	2	3	4	5
<i>Sales</i>							
Universe avg.			1.0	1.7	2.8	3.9	5.7
Sample avg.	2.2	2.0	1.0	2.0	3.0	4.5	6.6
Sample %			40.6	26.8	20.4	9.3	2.9
<i>Debt/assets</i>							
Universe avg.			0.0	0.0	0.2	0.3	1.0
Sample avg.	0.3	0.2	0.0	0.1	0.2	0.3	0.7
Sample %			19.8	17.2	16.3	22.7	24.1
<i>Short-term debt/total debt</i>							
Universe avg.			0.0	0.0	0.2	0.4	0.9
Sample avg.	0.3	0.1	0.0	0.0	0.2	0.4	0.9
Sample %			25.9	11.7	28.2	18.7	15.5
<i>Payout ratio</i>							
Universe avg.					0.0		55.6
Sample avg.	21.3	0.0			0.0		41.7
Sample %					50.2		49.8
<i>Credit rating</i>							
Universe avg.			4.7	5.8	6.8	7.3	8.4
Sample avg.	8.4	9.0	3.6	6.0	7.0	8.0	9.6
Sample %			7.5	8.4	7.8	16.5	59.8
<i>Historical growth</i>							
Universe avg.			-0.2	0.0	0.1	0.2	1.7
Sample avg.	0.3	0.1	-0.2	0.0	0.1	0.2	1.0
Sample %			1.0	26.6	33.3	25.2	13.9

Table A2

Public Sample Firms.

Comparison of the survey sample of publicly listed firms under management of US CEOs to the Compustat sample using survey variable definitions. Variable definitions as in Table A1.

	Sample average	Sample median	Compustat breakpoint quintiles				
			1	2	3	4	5
<i>Sales</i>							
Universe avg.			1.0	1.7	2.8	3.9	5.7
Sample avg.	3.5	3.0	1.0	2.0	3.0	4.7	6.6
Sample %			11.4	17.7	27.8	30.4	12.7
<i>Debt/assets</i>							
Universe avg.			0.0	0.0	0.2	0.3	1.0
Sample avg.	0.3	0.3	0.0	0.1	0.2	0.3	0.7
Sample %			18.8	18.8	11.6	30.4	20.3
<i>Short-term debt/total debt</i>							
Universe avg.			0.0	0.0	0.2	0.4	0.9
Sample avg.	0.2	0.2	0.0	0.0	0.2	0.4	0.9
Sample %			24.3	11.4	35.7	15.7	12.9
<i>Payout ratio</i>							
Universe avg. (%)					0.0		46.2
Sample avg. (%)	19.1	5.0			0.0		36.7
Sample %					53.2		46.8
<i>Credit rating</i>							
Universe avg.			4.7	5.8	6.8	7.3	8.4
Sample avg.	7.9	8.0	4.5	6.0	7.0	8.0	9.6
Sample %			7.7	17.3	13.5	23.1	38.5
<i>Historical growth</i>							
Universe avg.			-0.2	0.0	0.1	0.2	1.7
Sample avg.	0.2	0.1	-0.1	0.0	0.1	0.2	0.7
Sample %			3.0	36.7	24.1	26.6	10.1

(a) Table A1: Full survey sample versus Compustat

(b) Table A2: Public survey firms versus Compustat

7 Short summary

- The paper directly measures executive attitudes using survey and psychometric-style questions.
- CEOs are much more risk-tolerant and optimistic than the lay population.
- CEOs differ from CFOs in optimism, career path, age, tenure, education, and ownership.
- US executives are more risk-tolerant, more optimistic, and more patient than non-US executives.
- Optimistic CEOs are associated with higher debt usage and more short-term debt.
- Risk-tolerant CEOs are more likely to undertake acquisitions.
- Risk-tolerant, younger, and taller CEOs are more likely to be found in high-growth firms.
- Risk-averse CEOs receive less performance-sensitive pay and more salary.
- Impatient CEOs receive less stock/options/bonus compensation.
- The paper's main contribution is to show that directly measured managerial attitudes are related to corporate actions and compensation design.

8 Conclusion

8.1 Core conclusion

Graham, Harvey, and Puri show that managerial attitudes can be measured directly and that these attitudes are systematically related to corporate actions. CEOs are not merely generic agents implementing firm policy. They differ in optimism, risk tolerance, time preference, and sure-loss aversion, and these differences are associated with leverage, debt maturity, acquisition activity, firm growth, and executive compensation.

8.2 Economic intuition

The economic intuition is straightforward. Optimistic managers may expect favorable future outcomes and therefore be more willing to use short-term debt. Risk-tolerant managers are more willing to undertake acquisitions, which are risky and often require strong beliefs about future integration gains. Risk-averse managers dislike compensation packages that load heavily on stock, options, and bonuses because such pay exposes them to firm risk. Impatient managers discount future performance pay and therefore receive less performance-sensitive compensation.

8.3 Incremental contribution revisited

The paper advances behavioral corporate finance by moving from indirect proxies to direct measurement. It does not merely show that managers matter; it asks which attitudes managers have and how those attitudes relate to decisions. It also expands the empirical setting beyond public firms and links behavioral traits to compensation-contract design, a central prediction of agency theory.

8.4 Implications for corporate finance and governance

The findings imply that boards and investors should pay attention to executive attitudes when evaluating financing, investment, and acquisition strategies. A firm's capital structure or M&A behavior may partly reflect the preferences and beliefs of its CEO. Compensation committees should also recognize that incentive pay interacts with the executive's own risk tolerance and time preference.

8.5 Limitations

The study is observational and cross-sectional, so it cannot fully distinguish causation from matching. Executives may choose firms that fit their traits, or firms may hire executives whose traits fit the firm's policies. The survey is anonymous and self-reported, which limits external validation for many variables. Nonresponse and survey interpretation may introduce noise. The evidence should therefore be read as strong correlational evidence rather than definitive causal evidence.

8.6 Takeaway

The enduring takeaway is that managerial attitudes are economically meaningful. CEOs are unusually optimistic and risk-tolerant, and variation in those traits is related to how firms borrow, acquire, grow, and pay their leaders. The paper gives concrete empirical content to the idea that corporate policies partly reflect the people making the decisions.